

Technical Document #442: Data Engineering - Comprehensive Overview

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Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Stream processing handles continuous data streams in real-time.
- Partitioning divides data into smaller, more manageable pieces.
- Schema evolution handles changes in data structure over time.